

Clustering – Unit 5

What?

The purpose of clustering is to expand domain knowledge and provide meaningful inputs for machine learning models by uncovering hidden patterns within data. Clustering techniques group observations based on similarities and dissimilarities, enabling analysts to identify naturally occurring segments within a dataset.

Different similarity measures can be applied depending on the type of data. For example, the **Jaccard coefficient** is useful for asymmetric binary variables, where the presence of a characteristic is more important than its absence. In e-portfolio's [Unit 5](#), I calculated this coefficient for individuals from pathological test results. In addition, methods such as **k-means clustering** partition data into groups by minimizing within-cluster variation, while cluster centroids help describe the typical profile of each segment.

So What?

The concepts covered in this unit felt familiar because of my experience in the banking industry, where customer segmentation and recommender systems were used to target clients based on their profiles and behaviours. Clustering techniques played an important role in identifying customer groups with similar characteristics and tailoring products and marketing strategies accordingly.

What I found particularly interesting was the distinction between variables used to **define clusters** and those used to **explain or interpret them**. For example, k-means can be used to create clusters, while the resulting centroids can be analysed to build meaningful customer profiles and understand the key characteristics of each segment.

This unit reinforced the importance of combining statistical methods with business knowledge. Although clustering algorithms can generate mathematically valid groups, these groups may not always be practical or actionable. Without domain expertise, a model may identify patterns that have little strategic value or fail to align with real-world business objectives. Therefore, successful segmentation requires both technical rigor and a deep understanding of the context in which the analysis is applied.

Now What?

Going forward, I would like to develop a deeper understanding of how to evaluate clustering results and select the most appropriate clustering technique for different types of business problems.

Beyond creating clusters, I want to focus on assessing their stability, interpretability, and business relevance using metrics such as silhouette scores and validation techniques.

I also plan to explore more advanced clustering methods, such as hierarchical clustering and density-based approaches, to understand their advantages and limitations compared to k-means. In a professional setting, I expect this knowledge will help me design more effective customer segmentation strategies, improve recommendation systems, and generate insights that support data-driven decision-making. Most importantly, I will continue to emphasise the integration of analytical techniques with domain expertise to ensure that the insights produced are both statistically sound and practically valuable.